A proof of the conjectured closed form for OEIS A181280

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Abstract

Let $a(n)$ count the $4 \times n$ matrices M over $\mathbb{F}_2$ whose four rows, read as bit strings, are lexicographically increasing and whose Gram matrix $G = MM^\top \in \mathbb{F}_2^{4\times 4}$ has lexicographically decreasing rows. Kauers and Koutschan conjectured a closed form for $a(n)$, recorded as OEIS A181280 and still marked open. We prove their conjecture. Reading a matrix column by column gives a finite-state transfer matrix T with $a(n) = \mathbf{u}\, T^{\,n}\, \mathbf{e}_0$. Exact rational linear algebra yields an a priori bound on the recurrence order, and a Hankel computation determines the minimal recurrence of the tail ($n \geq 4$), whose characteristic polynomial is

$$(x-16)(x-8)^2(x+8)(x-4)^3(x+4)^2(x-2)^2.$$

The conjectured expression satisfies this recurrence and agrees with $a(n)$ on the required initial values, hence equals $a(n)$ for all $n \geq 4$. All computations use exact arithmetic and are supplied as ancillary code.

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1 Introduction

For a positive integer n, let M range over the $4 \times n$ matrices with entries in $\mathbb{F}_2 = \{0, 1\}$. Read each row of M as a length-n bit string, most significant bit first (column 1 is most significant), and order bit strings lexicographically. Let $G = G(M) = MM^\top \in \mathbb{F}_2^{4\times 4}$ be the Gram matrix over $\mathbb{F}_2$, i.e. $G_{ij} = \sum_{k=1}^{n} M_{ik}M_{jk} \bmod 2$; read each of its four rows as a 4-bit string, again most significant bit (column 1) first. Define

$$a(n) \;=\; \#\Big\{\, M \in \mathbb{F}_2^{4\times n} \;:\; \text{rows of } M \text{ strictly increasing, and rows of } G(M) \text{ strictly decreasing} \Big\}.$$

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This is the sequence **A181280** in the On-Line Encyclopedia of Integer Sequences (OEIS) [2], contributed by R. H. Hardin in 2010; it is row 4 of the table A181274. Its first nonzero values are $a(4) = 58$, $a(5) = 1629$, $a(6) = 28924$, $a(7) = 507052$, with $a(1) = a(2) = a(3) = 0$.

In their study of guessed-but-unproven recurrences in the OEIS, Kauers and Koutschan [1] subjected A181280 to their guessing pipeline. They recovered a plausible order-10 recurrence and the closed form below, but were unable to prove either; in their Table 1 the sequence carries the status letter "O" ("the case is open"), as opposed to "P" (recurrence proven) or "D" (D-finiteness proven) [1, §6.5]. The OEIS entry records the same expression with the parenthetical "(conjectured)".

Conjecture 1.1 (Kauers–Koutschan [1, Conjecture 20])**.** *For all $n \geq 4$,*

$$\begin{aligned} a(n) = {} & \tfrac{1}{3}\, 2^{\,2n-11}\big(6n^2 - 219n + 820\big) - \tfrac{1}{9}\, 2^{\,n-5}\big(3n + 32\big) - \tfrac{113}{3}\,(-1)^n\, 2^{\,3n-14} + 2^{\,4n-9} \\ & - \tfrac{1}{3}\,(-1)^n\, 2^{\,2n-11}\big(13n - 164\big) + \tfrac{1}{9}\, 2^{\,3n-14}\big(288n - 3473\big). \end{aligned} \tag{1}$$

We prove Conjecture 1.1.

Theorem 1.2. *Equation (1) holds for every integer $n \geq 4$.*

Proof strategy. Equation (1) was obtained from initial terms, so further term comparisons alone do not prove it; with an a priori bound of degree d on a candidate, agreement at $d+1$ points already determines it. We instead derive a recurrence independently from a finite-state model. Section 2 constructs the transfer matrix and establishes an a priori bound on the recurrence order, with no reference to (1). Section 3 determines the recurrence, proves that the conjectured formula satisfies it, and checks the required initial values.

Relation to prior work. The transfer-matrix argument is standard; see Stanley [4, Ch. 4] and Kauers–Paule [5, Ch. 4]. The contribution here is its application to A181280 and the resulting proof of Conjecture 1.1. The "Hardinian arrays" studied by Dougherty-Bliss and Kauers [3] form a different family (king-move arrays A253026/A253223/A253004) and do not include A181280; we did not find (1) proved elsewhere.

2 The transfer matrix and an a priori recurrence bound

We model the count by a deterministic-counting automaton that reads M one column at a time. A column is a vector $c \in \mathbb{F}_2^4$; there are 16 possible columns. After reading the first t columns, the only information about the prefix that can still affect acceptance is captured by a state defined now.

State. A state is a pair $s = (\mathrm{rel}, g)$ where:

- rel assigns to each of the $\binom{4}{2} = 6$ row pairs (i, j), $i < j$, a value in $\{$ `=`, `<`, `>` $\}$, recording the lexicographic comparison of the length-t prefixes of rows i and j: `=` if the prefixes are equal, `<` if row i's prefix is (strictly) smaller, `>` if larger. Because comparison is most-significant-bit first, once a pair is decided (`<` or `>`) it never changes; an `=` pair may later become `<` or `>`.

- $g \in \mathbb{F}_2^{4\times 4}$ (symmetric, 10 free bits) is the running Gram matrix $\sum_{k\le t} c^{(k)} c^{(k)^\top} \bmod 2$ of the columns read so far.

The initial state s_0 (empty prefix) has every pair `=` and $g = 0$.

Transition. Reading a column c updates a state (rel, g) to (rel', g') by:

- $g' = g + c\,c^\top \bmod 2$ (each $g'_{ij} = g_{ij} \oplus (c_i \wedge c_j)$);
- for each pair (i,j) with $\mathrm{rel}(i,j) =$ `=`: if $c_i \neq c_j$ the pair is now decided, becoming `<` if $c_i < c_j$ (that is $c_i = 0, c_j = 1$) and `>` otherwise; pairs already decided are unchanged.

We *prune* any state in which some pair (i,j), $i<j$, has become `>`: such a prefix already has row i exceeding row j, so the final matrix can never have all rows strictly increasing. Pruned states are dropped (they contribute 0 thereafter), which keeps the reachable state set finite and small.

Acceptance. A length-n run is *accepting* iff its terminal state (rel, g) has $\mathrm{rel}(i,j) =$ `<` for all $i<j$ (all rows strictly increasing, no ties left) *and* the four rows of the symmetric matrix g, read as 4-bit strings, are strictly decreasing.

Lemma 2.1. *$a(n)$ equals the number of length-n sequences of columns whose run from s_0 is accepting. Equivalently, let $\mathcal{S}$ be the (finite) set of reachable, unpruned states, $T \in \mathbb{Z}^{\mathcal{S}\times\mathcal{S}}$ the matrix with $T_{s',s} = \#\{c \in \mathbb{F}_2^4 : c \text{ sends } s \text{ to } s'\}$, $\mathbf{e}_0$ the indicator of s_0, and $\mathbf{u}$ the indicator (row) covector of accepting states. Then $a(n) = \mathbf{u}\,T^n\,\mathbf{e}_0$.*

Proof. The map sending M to its sequence of columns $(c^{(1)}, \dots, c^{(n)})$ is a bijection between $\mathbb{F}_2^{4\times n}$ and length-n column sequences. By induction on t, after reading $c^{(1)}, \dots, c^{(t)}$ the state (rel, g) equals exactly the pairwise prefix comparisons and the Gram matrix of the submatrix on the first t columns: the Gram update is the definition of G accumulated one column at a time ($MM^\top = \sum_k c^{(k)} c^{(k)^\top}$), and the comparison update is the definition of lexicographic order read most-significant-first. Hence the terminal state records precisely the data the two defining conditions test, so a run is accepting iff M is counted by $a(n)$; pruning only removes matrices already failing the increasing-rows condition. The matrix identity $a(n) = \mathbf{u}T^n\mathbf{e}_0$ is the standard expansion of the number of accepting walks: $(T^n\mathbf{e}_0)_s$ is the number of length-n column sequences whose run ends in state s, and $\mathbf{u}$ sums those over accepting s. □

The reachable, unpruned state set has $|\mathcal{S}| = 1242$ elements (enumerated by the breadth-first construction in the ancillary code). We never need T in closed form; we need only the following two consequences, both exact.

Lemma 2.2 (a-priori order bound)**.** *The sequence $\big(a(n)\big)_{n\ge 0}$ satisfies a linear recurrence with constant (integer) coefficients of order at most d_0, where $d_0 = \dim_{\mathbb{Q}} \operatorname{span}\{T^n\mathbf{e}_0 : n \ge 0\}$ is the dimension of the Krylov space of $\mathbf{e}_0$ under T. An exact rational computation gives $d_0 = 16$.*

Proof. Let $K = \operatorname{span}_{\mathbb{Q}}\{T^n\mathbf{e}_0 : n \ge 0\}$, a T-invariant subspace of $\mathbb{Q}^{\mathcal{S}}$ of dimension d_0. By construction K is the Krylov orbit of $\mathbf{e}_0$, so $\mathbf{e}_0$ is a *cyclic* vector for $T|_K$; hence $\deg \mu_K = \dim K = d_0$, where μ_K is the minimal polynomial of $T|_K$. The vectors $\mathbf{e}_0, T\mathbf{e}_0, \dots, T^{d_0}\mathbf{e}_0$ are $d_0 + 1$ elements of the d_0-dimensional space K, hence linearly dependent: there are rationals $\gamma_0, \dots, \gamma_{d_0}$, not all zero, with $\sum_{i=0}^{d_0} \gamma_i\, T^i\mathbf{e}_0 = 0$. (Take γ to be the coefficients of μ_K, which is monic of degree d_0, so $\gamma_{d_0} = 1$.) Applying $\mathbf{u}\, T^n$ on the left gives $\sum_{i=0}^{d_0} \gamma_i\, a(n+i) = 0$ for every $n \geq 0$. This is the asserted recurrence; its order is at most d_0.

The γ_i are in fact integers (though only rationality is needed below). The minimal polynomial μ_K of $T|_K$ divides the minimal polynomial of the integer matrix T, which in turn divides the (monic, integer) characteristic polynomial χ_T; so every root of μ_K is an algebraic integer, and a monic rational polynomial whose roots are algebraic integers has integer coefficients. The value $d_0 = 16$ is obtained by exact Gaussian elimination over $\mathbb{Q}$ on the Krylov iterates (ancillary `proof_matrix.py`). □

Lemma 2.2 bounds the recurrence order from the structure of the counting problem alone, using nothing about Equation (1). The remaining work is to identify the minimal recurrence within this bound and match the closed form to it.

3 The minimal recurrence and closed form

We work with the tail $\big(a(n)\big)_{n\geq 4}$ (the values $a(0) = \cdots = a(3) = 0$ are a boundary the closed form does not describe; see Remark 3.3). All computations below are in exact integer/rational arithmetic.

Lemma 3.1 (minimal recurrence)**.** *The minimal constant-coefficient linear recurrence satisfied by* $\big(a(n)\big)_{n\geq 4}$ *has order* 11*, with characteristic polynomial*

$$p(x) = (x-16)(x-8)^2(x+8)(x-4)^3(x+4)^2(x-2)^2. \tag{2}$$

Proof. Write $b_m = a(m+4)$ for the tail, and let E be the forward-shift operator $(Eb)_m = b_{m+1}$. We establish order ≥ 11 and order ≤ 11 separately, then read off p; the upper bound is where the a-priori bound of Lemma 2.2 does the work.

Lower bound (order ≥ 11*).* The Hankel matrix $H_{11} = \big(b_{i+j}\big)_{0\leq i,j\leq 10}$ is nonsingular: its determinant is a nonzero integer (computed exactly; ancillary `proof_matrix.py`). If the tail satisfied a monic recurrence of order $r < 11$, its shifts $b, Eb, \ldots, E^{10}b$ would be linearly dependent, forcing $\det H_{11} = 0$; hence no recurrence of order < 11 annihilates the tail.

Determination of p*.* Because H_{11} is nonsingular, a monic order-11 recurrence $\sum_{j=0}^{11} c_j\, b_{n+j} = 0$ ($c_{11} = 1$) has its coefficients $c_0, \ldots, c_{10}$ *uniquely* determined by the 11 shifted equations for $n = 0, \ldots, 10$, i.e. by the linear system $H_{11}\,(c_0, \ldots, c_{10})^\top = -(b_{11}, \ldots, b_{21})^\top$. Solving it exactly (ancillary `proof_matrix.py`) gives the monic polynomial

$$\begin{aligned} p(x) = x^{11} - 32x^{10} + 244x^9 + 1552x^8 - 27392x^7 + 70144x^6 \\ + 484352x^5 - 2781184x^4 + 1212416x^3 + 20185088x^2 - 48234496x + 33554432, \end{aligned}$$

with integer coefficients, which factors over $\mathbb{Z}$ as (2). (The same p is recovered independently in Section 4 by Berlekamp–Massey on a separately computed list of terms.) This p is so far only a candidate: the shifted system used only $b_0, \ldots, b_{21}$, so it expresses that $p(E)b_n = 0$ for $n = 0, \ldots, 10$, not yet for all n.

Upper bound (order ≤ 11*), via the a-priori recurrence.* By Lemma 2.2 there is a monic polynomial q with $\deg q = d_0 \leq 16$ and $q(E)a = 0$ on all of $\mathbb{N}$; hence $q(E)b = 0$ as well. Consider the residual sequence $r = p(E)b$. Since shift-operator polynomials commute,

$$q(E)\,r \;=\; q(E)p(E)\,b \;=\; p(E)q(E)\,b \;=\; p(E)\,0 \;=\; 0,$$

so r itself satisfies the monic order-d_0 recurrence $q(E)r = 0$. A monic recurrence of order d_0 propagates: if $r_n = 0$ for d_0 consecutive indices $n = n_0, \dots, n_0 + d_0 - 1$, then $r_{n_0+d_0} = 0$, and inductively $r_n = 0$ for all $n \geq n_0$. An exact computation shows $r_n = p(E)b_n = 0$ for the 26 consecutive indices $n = 0, \dots, 25$ (equivalently $p(E)a(m) = 0$ for $m = 4, \dots, 29$; ancillary `proof_matrix.py`); since $26 \geq d_0$, the propagation gives $r_n = 0$ for *all* $n \geq 0$. Thus $p(E)$ annihilates the entire tail, so the tail satisfies a recurrence of order ≤ 11.

Combining the two bounds, the minimal constant-coefficient recurrence of $\big(a(n)\big)_{n\geq 4}$ has order exactly 11 and characteristic polynomial $p =$ (2). (The residual is nonzero at $n = 0, 1, 2, 3$, i.e. $p(E)a(m) \neq 0$ for $m = 0, 1, 2, 3$, consistent with the boundary at $m = 4$ of Remark 3.3.) $\square$

Lemma 3.2 (the closed form solves the same recurrence)**.** *Let $C(n)$ denote the right-hand side of* (1)*. Then C satisfies a constant-coefficient linear recurrence whose characteristic polynomial is exactly $p(x)$ of* (2)*; in particular $\sum_{i=0}^{11} p_i\, C(n+i) = 0$ for all integers n.*

Proof. Expanding (1) and grouping by exponential type writes $C(n)$ in the form $\sum_i P_i(n)\,\gamma_i^n$ of Stanley [4, Thm. 4.1.1(iii)], namely as a $\mathbb{Q}$-linear combination of the eleven sequences

$$16^n, \quad 8^n,\ n\,8^n, \quad (-8)^n, \quad 4^n,\ n\,4^n,\ n^2 4^n, \quad (-4)^n,\ n\,(-4)^n, \quad 2^n,\ n\,2^n,$$

with (nonzero) coefficients $\frac{1}{512}, -\frac{3473}{147456}, \frac{1}{512}, -\frac{113}{49152}, \frac{205}{1536}, -\frac{73}{2048}, \frac{1}{1024}, \frac{41}{1536}, -\frac{13}{6144}, -\frac{1}{9}, -\frac{1}{96}$ respectively (ancillary `derive_closed_form.py`). In the notation of [4, Thm. 4.1.1], C has the form (4.3) with distinct nonzero bases $\gamma_i \in \{16, 8, -8, 4, -4, 2\}$ and polynomial parts P_i of degrees $0, 1, 0, 2, 1, 1$; by the theorem's equivalence (iii)$\Leftrightarrow$(ii) the denominator is $\prod_i (1 - \gamma_i x)^{d_i}$ with $d_i = \deg P_i + 1$, i.e. multiplicities $16 : 1$, $8 : 2$, $-8 : 1$, $4 : 3$, $-4 : 2$, $2 : 2$. Reversing each factor $(1 - \gamma x) \mapsto (x - \gamma)$ gives exactly the characteristic polynomial (2), and C satisfies its associated recurrence (4.2). (One also checks the eleven annihilations directly in the ancillary code: $n^k \gamma^n$ is killed by $(\text{shift} - \gamma)^{k+1}$, hence by any shift-polynomial with γ a root of multiplicity $> k$.) $\square$

Proof of Theorem 1.2. By Lemmas 3.1 and 3.2, both $\big(a(n)\big)_{n\geq 4}$ and $\big(C(n)\big)_{n\geq 4}$ satisfy the order-11 linear recurrence with characteristic polynomial p, i.e. $\sum_{i=0}^{11} p_i\, x(n+i) = 0$ for $x \in \{a, C\}$ and all $n \geq 4$. A constant-coefficient linear recurrence of order 11 with invertible leading coefficient ($p_{11} = 1$) determines each term from the preceding 11 (Stanley [4, Thm. 4.1.1]: in the equivalence there, condition (ii) is exactly such a recurrence, and "we may choose $f(0), \dots, f(d-1)$ and then the other $f(n)$'s are uniquely determined"); by a trivial induction, two solutions that agree on any 11 consecutive indices agree everywhere on the domain of the recurrence. The two sequences agree at the 11 consecutive indices $n = 4, 5, \dots, 14$:

$a(4) = 58$,	$a(5) = 1629$,	$a(6) = 28924$,	$a(7) = 507052$,
$a(8) = 8211776$,	$a(9) = 133693904$,	$a(10) = 2140571200$,	$a(11) = 34361115072$,
$a(12) = 549587348992$,	$a(13) = 8798356254976$,	$a(14) = 140744002571264$,	

each equal to $C(n)$ in exact arithmetic (ancillary; the seed values are also listed in `code/seeds.txt`). Therefore $a(n) = C(n)$ for all $n \geq 4$, which is Theorem 1.2. $\square$

Remark 3.3 (the $n \geq 4$ boundary is genuine)**.** For $n < 4$ there are no valid matrices ($a(1) = a(2) = a(3) = 0$, and $a(0) = 0$ vacuously), whereas $C(0), \dots, C(3)$ are the non-integers $\frac{103}{4096}, -\frac{71}{1024}, -\frac{1}{64}, -\frac{1}{16}$. The closed form is the count only on $n \geq 4$, exactly as Conjecture 1.1 states; the recurrence likewise annihilates the sequence only from $n = 4$ onward.

Remark 3.4 (order 11 versus the guessed order 10)**.** Kauers–Koutschan report a guessed recurrence of order $r = 10$ [1, Table 1]. There is no conflict: their order counts a *P-recursive* (polynomial-coefficient) recurrence, a strictly more general and typically lower-order normal form, whereas our 11 is the order of the minimal *constant*-coefficient (C-finite) recurrence, equal to the degree of the characteristic polynomial, i.e. the sum $1+2+1+3+2+2 = 11$ of the root multiplicities. Both descriptions are correct for the same sequence.

4 Computational verification

The proof uses three exact computations: the dimension of the Krylov space of $(T, \mathbf{e}_0)$, the determinant and shifted solve associated with H_{11}, and the 26 recurrence residuals of Lemma 3.1. The ancillary scripts reconstruct these objects from the transition rules of Section 2 using integer or rational arithmetic; no floating-point calculations are used. Concretely:

1. the Krylov dimension of $(T, \mathbf{e}_0)$ is 16 (Lemma 2.2);
2. the order-11 Hankel matrix H_{11} of the tail is nonsingular (its determinant is a nonzero integer), and the shifted 11×11 Hankel system yields the integer-coefficient p of (2);
3. the residual $p(E)a(n)$ vanishes for the 26 consecutive indices $n = 4, \ldots, 29$, at least the $d_0 \le 16$ the propagation argument of Lemma 3.1 requires, and is nonzero for $n = 0, 1, 2, 3$.

These are performed in `proof_matrix.py`. The reachable state set has 1242 elements ($a(n) = \mathbf{u}T^n\mathbf{e}_0$), and `code/README.md` maps each computation to its driver and expected output; `code/MANIFEST.sha256` lists file hashes. Every object is rebuilt from the definitions, with no external data file.

Independent checks. Several further computations, not needed for the proof, corroborate it. The transfer matrix reproduces the OEIS values for $n = 1, \ldots, 12$ (`transfer.py`); two independent brute-force implementations agree for $n = 4, 5, 6$, and the row-tuple implementation also verifies $a(7) = 507052$ (`brute.py`, `brute_indep.py`); the polynomial p is recovered again by Berlekamp–Massey on a separately computed term list, and the exponential-basis solution of Lemma 3.2 agrees with (1) symbolically (`derive_closed_form.py`); and $a(n) = C(n)$ for $n = 4, \ldots, 60$ as an additional check (`recurrence.py`).

A separate route recomputes $a(n)$ without the automaton of Section 2: `machine_fourier_coupled.py` applies a Walsh–Hadamard (Fourier) diagonalization over $(\mathbb{Z}/2)^{10}$ that decouples the Gram constraint from the ordering constraint, reducing the live state space to 8 ordering states recombined by Fourier inversion against a fixed weight over the 48 strictly-decreasing Gram matrices. This different state space reproduces $a(4), \ldots, a(12)$, the characteristic polynomial (2), and the closed form (1).

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